

The distributions listed below are directly available in Microsoft Store:

- Ubuntu 16.04 LTS / 18.04 LTS / 20.04 LTS
- openSUSE Leap 15.1
- SUSE Linux Enterprise Server 12 SP5 / 15 SP1
- Kali Linux
- Debian GNU/Linux
- Fedora Remix for WSL
- Pengwin
- Pengwin Enterprise
- Alpine WSL

(There are many other Linux distributions that are unofficially supported by WSL.)

For Linux application developers, WSL provides the Linux development environment — which comprises most of the command line tools, utilities and applications — directly on Windows without the overhead of a traditional virtual machine or dual-boot machine setup. This is a far better user experience than Cygwin, which was used for basic Linux development environments on the Windows OS.



Note: Microsoft released WSL2 in 2020, which is an improved version and faster than the first version. However, for simplicity we are considering the first version of WSL in this article.

The prerequisites for WSL are:

- Windows 10 (version 1607 or above).
- WSL only runs on 64-bit versions of Windows 10. The 32-bit version is not supported.

Installation

Here is an example of setting up a WSL environment using Ubuntu 18.04 LTS. There are two ways to do this — through Microsoft Store and through the PowerShell command line.

Microsoft Store: You can directly install the desired distro by searching for it (say, Ubuntu) in Microsoft Store. If Microsoft Store is not installed or not

accessible, you can install it through the command line as mentioned next.

PowerShell: Open PowerShell as administrator and follow the steps given below.

1. The first step is to enable the *Windows optional feature—WSL* by executing the command given below:

```
Enable-WindowsOptionalFeature -Online
-FeatureName Microsoft-Windows-
Subsystem-Linux
```

It will ask for a computer restart.

2. Once the machine has restarted, the next step is to download Ubuntu-18.04 through the PowerShell command line (for a different version, kindly check <https://docs.microsoft.com/en-us/windows/wsl/install-manual>):

```
Invoke-WebRequest -Uri https://aka.ms/
wsl-ubuntu-1804 -OutFile Ubuntu.appx -
UseBasicParsing
```

3. Extract the downloaded distribution by executing the following commands:

```
Rename-Item -Path ~/Ubuntu.appx
-NewName Ubuntu.zip
Expand-Archive ~/Ubuntu.zip ~/Ubuntu
```

4. To run the Linux distro (Ubuntu), change to the extracted directory and execute the binary:

```
cd Ubuntu
./ubuntu1804.exe
```

5. Once you run the distro name executable, it will start setting up the environment and ask for a new user name and password; after that you're good to go.

Leveraging Microsoft development tools

As WSL is natively integrated with the Windows OS, developers can

build and debug Linux applications with Windows tools like Visual Studio Code and Visual Studio. Visual Studio Code (<https://code.visualstudio.com>) is a free and powerful IDE, which has an extension that allows you to attach VSCode to WSL directly (refer to the document explaining this process at <https://code.visualstudio.com/docs/remote/wsl>). This helps developers to use the powers of the GUI IDE of Windows and command line of Linux simultaneously. There are many other extensions (code formatting, Git history, Docker, etc) in VSCode that give developers lots of flexibility.

WSL vs virtual machine software

Performance is always a concern for developers when they use VMs. Most benchmarks suggest that WSL performance is nearly as good as native Linux distributions. However, in some I/O tests bottlenecks were experienced while using WSL. The new version, WSL2, has improved upon this.

WSL has quite a few benefits over virtual machine software. These are:

- It's lighter than a virtual machine and consumes fewer resources (CPU, memory and storage).
- Easy to setup.
- Natively integrated with Windows OS and supported by Microsoft.
- No third party VM software is required.
- Fairly large community and good documentation available from Microsoft, which allows even developers who have never used Linux before to get started easily.
- Speeds up the development process in case of cross-platform projects.
- You can share Windows apps and Linux tools on the same set of files.
- In general, almost near native Linux distro performance.
- Very stable.
- Cost-efficient (no licence fee).
- Users can install the required packages (for example, if you are running the WSL Ubuntu image,